A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 05

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**Marks shared** Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. **up to 2 Organic advantages:** often waste products, locally available and cheap – add humus improving structure and water retention – increase soil biota – slow release reduces leaching **up to 2 Organic disadvantages:** nutrient composition cannot be controlled – released slowly so needs long-term planning – bulky with high water content so expensive to transport – cannot be applied to a growing crop – odour / pathogens **up to 2 Inorganic advantages:** composition controlled to match crop needs – nutrients released rapidly – concentrated and easy to apply **up to 2 Inorganic disadvantages:** high energy (fossil fuel) manufacture (Haber process) – no organic matter so humus and biota decline – some toxic to worms – high solubility so leached causing eutrophication – raw materials (phosphate rock) finite **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	6
01.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Crops grown in a circulating nutrient solution (inert medium) in greenhouses – nutrient concentrations monitored and adjusted so nutrients are never limiting – abiotic factors controlled **up to 2** Roots are small (all in contact with nutrients) so more growth goes into the harvested crop – no soil-borne pathogens or weeds – crop stays fresh with roots **up to 2 Disadvantages:** high inputs of nutrients and energy (heating, lighting, pumping) – high level of technical knowledge – only economic for high-value crops **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Less dead organic matter / leaf litter so less humus and slower soil formation – fewer nutrients returned **1** Loss of canopy and litter exposes soil to rain splash and wind **1** Decomposing roots no longer bind soil so erosion, gullying and landslides increase (especially on slopes) **1** Reduced organic matter and biota lower infiltration and water retention **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	4
02.2	**1 mark each** 1 mark for each of the 4 points below. **1** Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind **1 Increased runoff:** more flooding after rain, lower flows in dry periods – sedimentation of rivers **1** Increased albedo of cleared land (reflects more sunlight) **1** Reduced carbon sequestration and loss of the carbon reservoir (CO_2 released by burning / decay) contributing to climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
02.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects **1** Loss of inter-species relationships (pollinators, seed dispersers) causing further declines **1** Loss of genetic resources / potential medicines before they are discovered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis – human processes usually use high energy-density non-renewable fossil fuels **1** Natural processes occur at low temperatures because enzymes lower the activation energy of reactions – human processes need high temperatures and pressures **1** Natural processes produce no toxic wastes / their products are raw materials for other processes – industrial processes emit CO2 and NOx **1** So natural fixation is sustainable indefinitely with a low carbon footprint **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
03.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Hydrological cycle (evaporation, precipitation) **1** Carbon cycle / photosynthesis **1** Atmospheric circulation / winds **1** Thermohaline circulation (density differences from heating / evaporation) **1** Decomposition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Thermohaline** → the ocean conveyor driven by temperature and salinity **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration	3

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting **1** Anaerobic digestion and composting treat wastes – enzymes in industry (biological washing powders, biofuels from algae / microorganisms) **1** Heat pumps and solar thermal capture low-density heat – low-temperature manufacturing using biological polymers – fuel cells instead of combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Coordinating global data on biodiversity conservation (Red List) **1** Increasing understanding of the importance of biodiversity **1** Deploying nature-based solutions to global challenges in climate, food and sustainable development **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Decline = (12 000 - 2100) / 12 000 x 100 = 82.5% **1** Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km^2) and adults (2100) only meet Endangered **1** A species is placed in the highest category for which it meets any one criterion, so Critically Endangered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Categorising a species requires a great deal of research (population, range, trends) and resources – about 85 000 of millions of species assessed **1** Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) **1** Evolutionarily unique species / EDGE species – endemic species – keystone species – flagship species – species with fragmented populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **endemic** → found naturally only in that one place • **keystone species** → a species whose loss changes the whole community	3
04.4	1 mark for the point below. **1 Iberian lynx:** Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years) – Fregate Island beetle after rat eradication **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	
05.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
06.1	1 mark for the point below. **1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Zebu tolerate heat but have a low milk yield – Ayrshire have a high milk yield but are not heat tolerant **1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
06.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Replacement of valuable animals that die with genetically identical individuals **1** Replacement of herds culled during a disease outbreak **1** Production of large numbers of individuals with desirable characteristics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1
07.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 2 400 / 2 000 **1** = 1.2 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **food conversion ratio** → the feed needed per unit of food produced **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Carnivorous species have a high market value, so they are the most profitable to farm, and consumers in wealthy countries prefer them **1** They are fed on fishmeal and fish oil made from wild-caught fish, so farming them still depends on catching wild fish and does not take the pressure off marine stocks **1** Several tonnes of wild fish may be needed per tonne of salmon, so the extra trophic level means the total amount of fish protein available falls rather than rises **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
08.3	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Uneaten feed and faeces fall through the cages and accumulate on the sea bed, where their decomposition removes oxygen and smothers the benthic community **1** The nutrients released cause local eutrophication and can trigger algal blooms **1** Parasites, especially sea lice, build up in the dense stock and spread to wild salmon and sea trout passing the cages **1** Antibiotics, pesticides used to treat sea lice and antifouling chemicals from the nets disperse into the water and affect non-target species **1** Escaped farmed fish interbreed with wild populations, diluting the locally adapted gene pool, and compete with them for food and spawning sites **1** Predators such as seals and cormorants attracted to the cages are shot or trapped **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4

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Question	Marking guidance	Marks
08.4	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Integrated multi-trophic aquaculture:** seaweeds and filter feeders such as mussels are grown alongside the cages and take up the dissolved nutrients and particulate waste as their own food **1** Siting cages where the current is strong, and leaving sites fallow between cycles, so wastes are dispersed and the sea bed recovers **1** Lowering stocking density and vaccinating the fish, so less medication is needed, and using cleaner fish such as wrasse to remove sea lice instead of pesticides **1** Stronger, better maintained nets and predator exclusion nets, so fewer fish escape and predators do not have to be killed **1** Replacing part of the fishmeal in the feed with plant or insect protein, reducing the demand on wild fish stocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
08.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Farming carnivorous species still depends on wild fish for feed, so expanding it increases rather than reduces the catch of wild fish **1** Suitable sheltered coastal sites are limited, and they compete with other uses and with sensitive habitats such as mangroves, which shrimp farming has already destroyed on a large scale **1** Some valuable species, such as tuna, cannot yet be bred reliably in captivity, so farms are stocked with wild-caught young fish and the wild population is still exploited **1** The local pollution, disease and escape problems mean that simply scaling the industry up multiplies its impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-359; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Warming melts ice shelves that hold back glaciers and lubricates ice flow so land ice reaches the sea faster (Antarctic Peninsula most affected) – sea ice loss removes krill habitat and lowers albedo **1** Raised UV harms planktonic algae, krill and fish / sea urchin larvae in surface waters (land animals protected by fur / feathers) **1** Disturbance of wildlife, introduction of non-indigenous species and pathogens, fuel and sewage pollution – visitor numbers rising rapidly at a few landing sites **1** Reserves of coal, oil, gold, uranium, platinum etc – pressure to exploit will rise as reserves elsewhere deplete though currently banned and operations extremely difficult **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
09.2	**1 mark each** 1 mark for each of the 6 points below. **1** Treaty signed by 52 countries including all active there – no military activity (except supporting research), no nuclear explosions or radioactive waste dumping **1** No commercial mineral exploitation **1 Fisheries regulated by CCAMLR:** krill and some fish exploited, fisheries closed if over-exploited – no commercial exploitation of mammals **1 Tourism control:** no accommodation on land (wastes stay on ships), one ship per site with maximum 100 ashore, trained guides, keep 5 m from wildlife, no litter, cleaned protective clothing to prevent introductions **1 Waste management:** all wastes removed (fuel containers, batteries, plastics, oil) – sewage only discharged where sea currents disperse it – old waste removed unless removal causes more damage **1** Only long-term residents are research scientists and support staff – strict regulation of research impacts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	6
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 69-71; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
10.2	**1 mark each** 1 mark for each of the 4 points below. **1** Negative feedback resists change / returns the system towards its previous state **1** eg increased temperature increases cloud cover and albedo, or higher CO_2 increases photosynthesis **1** Positive feedback increases the change / makes it self-sustaining **1** eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
10.3	1 mark for the point below. **1** Many species with a complex food web so the loss of one has a small effect – agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles of natural systems:** dynamic equilibria and negative feedback – diverse systems resistant to change – low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles) – low-temperature processes using enzymes – material cycles with wastes as raw materials – small number of abundant elements – biodegradable non-toxic wastes • **Human systems:** non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) • **Applications:** renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy • Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires) – tipping points • **Barriers:** population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure • Justified conclusion on how far natural principles can be applied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **biodegradable** → broken down by decomposers **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Natural systems have remained sustainable because they run on low energy-density renewable energy, mainly sunlight driving the hydrological, carbon and nitrogen cycles; their reactions occur at low temperatures because enzymes lower activation energies; materials cycle so the waste of one process is the raw material of the next; only a few abundant elements are used; and wastes are biodegradable and non-toxic. Negative feedback keeps them in dynamic equilibrium and high diversity makes them resistant to change.	
	Human systems contrast on every point: non-renewable fossil fuels, high-temperature processes such as the Haber process and incineration, linear use that disperses minerals and mixes materials that cannot be separated, and persistent toxins like PCBs and the metals in electronics. We also override negative feedback - when degraded soil reduces yields we add fertiliser, irrigation and pesticides - and trigger positive feedback such as permafrost melting.	
	Copying natural principles is already possible in many areas: renewable energy and heat pumps, bioleaching at low temperature, recycling and design for disassembly, industrial symbiosis, legumes and organic matter instead of manufactured nitrate, and retaining hedgerows for pest predators - the basis of the circular economy. The limits are population size and affluence, the attitude of urban consumers who never see depleted resources, cost, and the need for international cooperation. So the principles can make individual activities far more sustainable, but only a large-scale, cooperative shift in lifestyle would make the whole human system sustainable.	
	Weaker answer - Level 2 (5 marks)	
	Natural systems use renewable energy from the Sun and recycle materials, so nothing runs out and there is no waste. Humans use fossil fuels which will run out and produce waste that is toxic. Natural processes happen at low temperatures but the Haber process needs high temperatures. Humans could copy natural systems by using renewable energy like solar and wind, recycling more and using less fertiliser. This would make human activities more sustainable. However it would be expensive and people would have to change their lifestyle.	
	What makes the difference	
	• Level 2 states the contrast correctly but does not explain the mechanisms (enzymes, negative feedback, diversity, dispersal of minerals) that Level 3 uses.	
	• Level 3 applies the principles to specific technologies (bioleaching, heat pumps, industrial symbiosis, legumes); Level 2 stays at 'recycle more'.	
	• 'Evaluate the extent' - Level 3 distinguishes individual activities from the whole system in its conclusion; Level 2's evaluation is a single sentence about cost.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Resources:** wood, fibres, oils, fuels, food – new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • **Biomimetics:** wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • **Medicines:** poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks – few species yet studied – physiological research (marsupials, hippo, sponges) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools and inbreeding • **Ecosystem services:** atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance – interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • **Evaluation:** economic vs intrinsic / moral arguments – services unnoticed because of dynamic equilibrium – unknown future value – justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

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